Golla Shriram

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.97
Sri Chaitanya Junior Kalasala	Class XII (TSBIE)	2024	96.10%
Sri Chaitanya Techno School	Class X (SSC)	2022	10.00%

SCHOLASTIC ACHIEVEMENTS

- Qualified for **INChO (2nd level)** through NSEC - National Chemistry Olympiad

RESEARCH EXPERIENCE

Machine Unlearning Research : Working under Dr. Konda Reddy Mopuri

- Involved in Ongoing Research on Machine Unlearning mainly in Vision Settings i.e Discriminative and Generative Machine Unlearning
- Currently exploring methods for Approximate Unlearning in Discriminative setting like class wise unlearning and instance wise unlearning

PROJECTS

Handwritten Notes Knowledge Base with RAG (Link)

- Built an end to end pipeline which transcribes and summarizes handwritten notes using VLMs and stores them as Markdown.
- Implemented contextual retrieval to improve retrieval accuracy on short, ambiguous notes.
- Built a self-reinforcing knowledge base: conversations are automatically summarized via LLM and are updated in Knowledge Base

DCGAN for Face Generation (Link) : Synthetic Face Generator

- Implemented DCGAN from scratch in PyTorch
- Trained on 200 K images of CelebA for 20 epochs using Adam optimizer and used weight initializations, real-fake batch normalizations to improve training stability of GAN's.

Mini-ML Library : (Link)

- Contains several generative models like MLE, MAP, Gaussian Naive Bayes, BNB , LDA etc
- Implemented an **Autograd Engine** which supports Automatic differentiation with back-propagation using Numpy
- Supports **Computational Graph visualization** by using Graphviz.
- Ran Experiments on **FashionMNIST** achieving a competitive performance with scikit-learn library in terms of accuracy.
- Implemented Unsupervised models like PCA,K means clustering , EM based methods .

Neural Networks from Scratch : (Link)

- Built customizable MLP which supports Backpropagation (Uses custom optimizer) from scratch using Numpy .
- Supports custom loss functions like Huber loss , activations like LeakyReLU, regularizations and weight initializations like **Xavier and He.**
- Implemented various kinds of optimizers like **Adam, AdaGrad, Muon, RMSProp, Nestrov AG** from scratch.
- Ran Ablations using varying depth, number of layers, loss functions, optimizers , running time on a Custom Generated Dataset.
- Compared performance by running a SVR using different kernels like **Linear, Polynomial, RBF and Neural Kernel**

Fair Representation Learning & Graph Neural Networks (Link)

- Explored fair representation learning for high-dimensional data using **Variational Inference** and **Information Bottleneck theory**
- Explored **invariant representations** by using **Flexible Fair VAEs (FFVAE)** to disentangle sensitive attributes.
- Explored Graph Neural Networks and **Fairness in Graph Neural Networks**.
- Implemented the VFAE, FFVAE, and Multi-view Information Bottleneck (MIB) framework .

RELEVANT COURSES

- **AI :** Foundations of Machine Learning, Deep Learning, Optimization-II, Optimization-I, AI for Humanity, Concentration Inequalities
- **Mathematics :** Numerical Analysis, Probability and Random Variables, Series of Functions, Calculus - I & II, Differential Equations, Intro to Metric Spaces
- **Computer Science :** Algorithms, Data Structures & Applications, DBMS-I, OS-I , Discrete Maths
- **Electrical Engineering :** Linear Systems & Signal Processing, Matrix Theory, Control Systems, Random Processes

SKILLS

- **Languages***:* Python, C++, C, SQL, LaTeX, Git
- **Frameworks***:* PyTorch, Hugging Face, FastApi, Scikit-learn, NumPy, Matplotlib, Pandas, BeautifulSoup

POSITIONS OF RESPONSIBILITY

Matrix Theory - Undergraduate Teaching Assistant under *Dr. GVV Sharma*

- Conducted Viva's and Corrected Assignments for undergraduate coursework

Machine Learning Core in Tinkerer's Lab

- Conducted Sessions for people to get interested in Machine Learning